Robot Dynamics & Control

2026

Exercises for Exam

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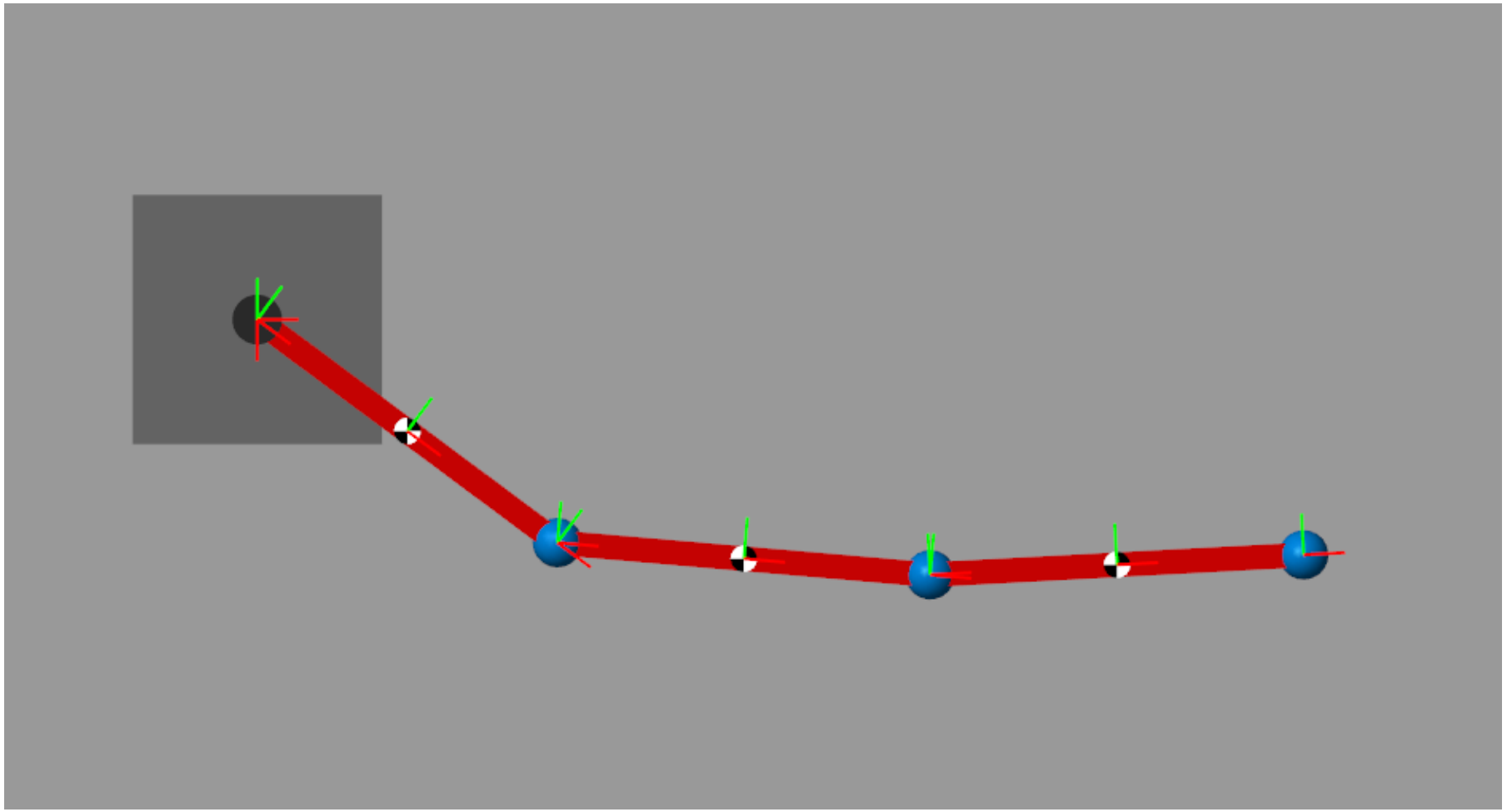
WARNING

The following exercise suggestions are variations on topics related to the previous exercise sessions

Before working on the following exercises you should have completed successfully the exercises presented during the course

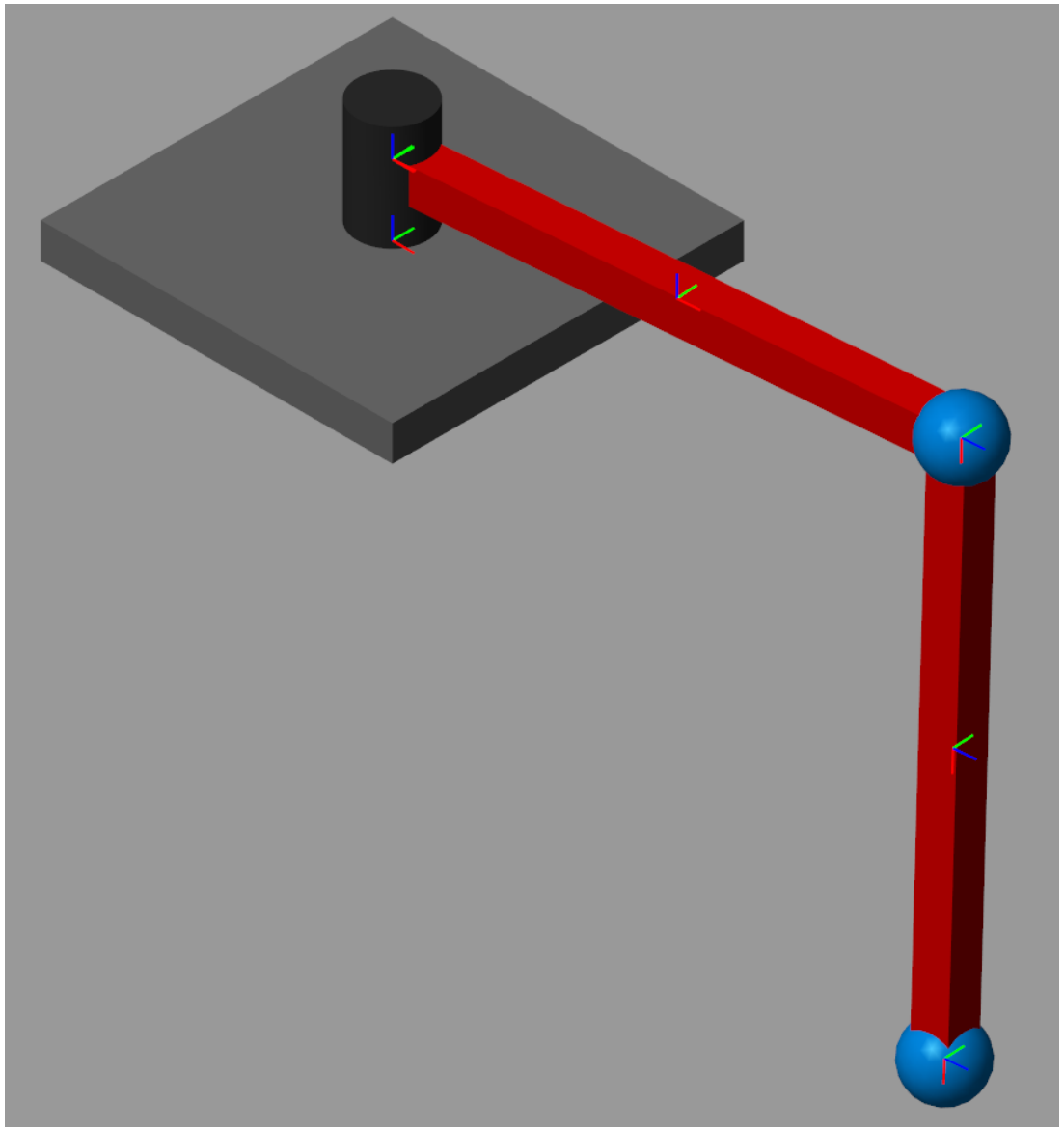
Try different models

A 3R planar robot



Try different models

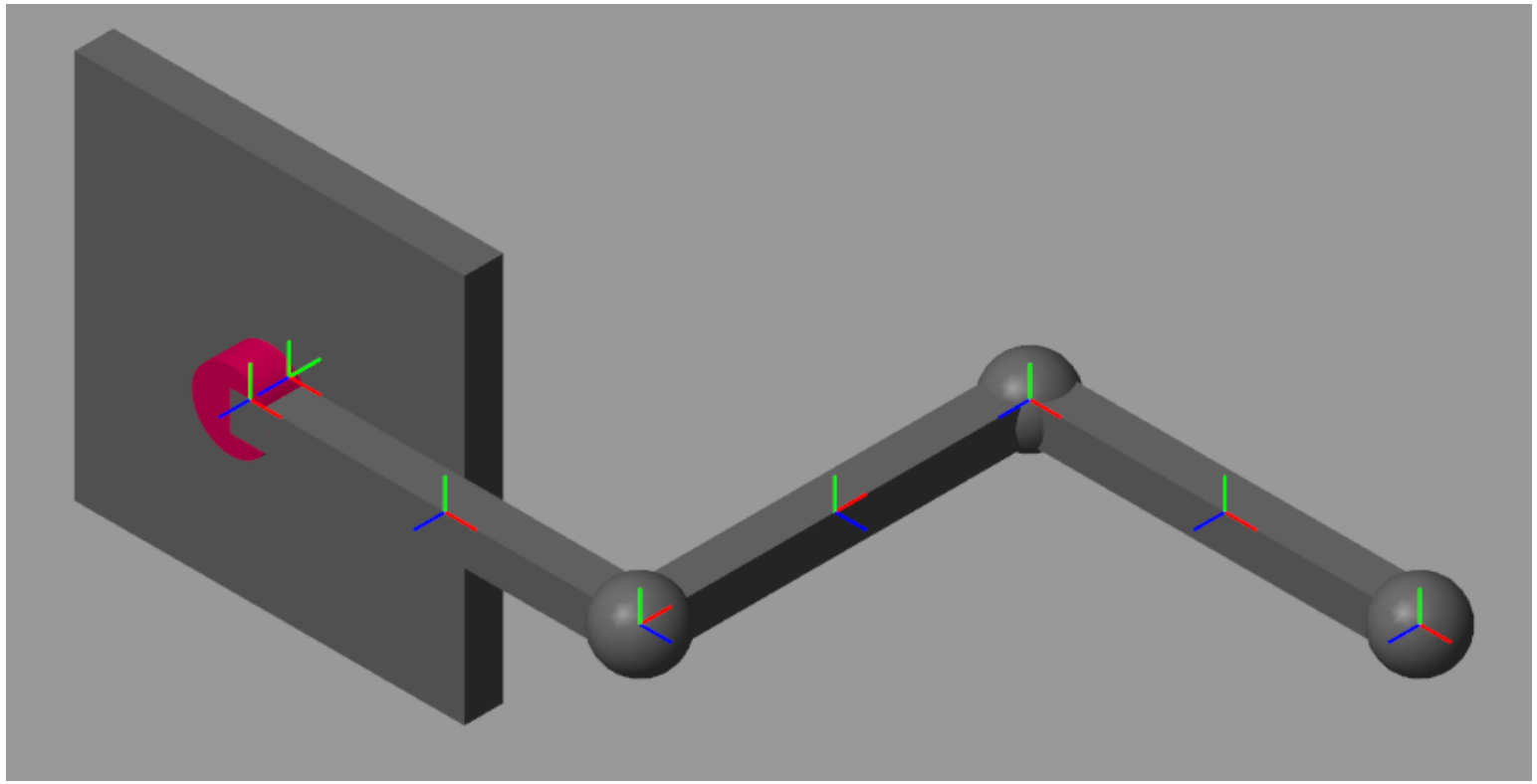
A 2R non-planar robot



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Try different models

A 3R non-planar robot



Exercises

- Implement **static** modeling, **dynamic** modeling and dynamic **control** (PD + gravity and Computed Torque) with **different robotic structures**

- Make **variations** to the structures: change the **length** or the **mass** of one or both links, **add objects** to the basic structure and consider them for dynamic modeling (Implement **static** modeling and PD+gravity **control**)

